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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import joblib
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

# Loading the dataset
# Assuming columns: ['floors', 'lateral_force_N', 'max_stress_MPa', 'max_displacement_mm', 'safety_factor_min']
df = pd.read_csv('/content/dataset_quakeguard_corrected.csv')

# Define inputs (X) and outputs (y)
X = df[['floors', 'lateral_force_N']]
y = df[['max_stress_MPa', 'max_displacement_mm', 'safety_factor_min']]

# Splitting into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# exploratory data analysis
plt.figure(figsize=(15, 5))

# Plots
# Safety Factor vs Load
plt.subplot(1, 3, 1)
sns.scatterplot(data=df, x='lateral_force_N', y='safety_factor_min', hue='floors', palette='viridis')
plt.title('Safety Factor vs. Applied Load')
plt.xlabel('Lateral Force (N)')
plt.ylabel('Min Safety Factor')

# Stress vs Load
plt.subplot(1, 3, 2)
sns.scatterplot(data=df, x='lateral_force_N', y='max_stress_MPa', hue='floors', palette='viridis')
plt.title('Max Stress vs. Applied Load')
plt.xlabel('Lateral Force (N)')
plt.ylabel('Max Stress (MPa)')

# Displacement vs Load
plt.subplot(1, 3, 3)
sns.scatterplot(data=df, x='lateral_force_N', y='max_displacement_mm', hue='floors', palette='viridis')
plt.title('Max Displacement vs. Applied Load')
plt.xlabel('Lateral Force (N)')
plt.ylabel('Max Displacement (mm)')

plt.tight_layout()
plt.show()

# Model Training (With Target Transformation)

# 1. Transform the training target for Safety Factor to make it linear (1 / SF)
y_train_transformed = y_train.copy()
y_train_transformed['safety_factor_min'] = 1 / y_train['safety_factor_min']

# 2. Train the Linear Regression model on the transformed data
model = LinearRegression()
model.fit(X_train, y_train_transformed)

joblib.dump(model, 'structural_ml_model_linear.pkl')
print("Model trained and saved as 'structural_ml_model_linear.pkl'\n")

# Model Evaluation
y_pred_transformed = model.predict(X_test)

# Inverting the safety factor predictions BACK to their original scale
y_pred = y_pred_transformed.copy()
y_pred[:, 2] = 1 / y_pred_transformed[:, 2]

# Calculating metrics against the ORIGINAL, unmodified y_test
r2 = r2_score(y_test, y_pred, multioutput='raw_values')
mae = mean_absolute_error(y_test, y_pred, multioutput='raw_values')

print("--- Model Evaluation (After Transformation Fix) ---")
print(f"Max Stress:          R² = {r2[0]:.4f}, MAE = {mae[0]:.4f} MPa")
print(f"Max Displacement:    R² = {r2[1]:.4f}, MAE = {mae[1]:.4f} mm")
print(f"Min Safety Factor:   R² = {r2[2]:.4f}, MAE = {mae[2]:.4f}\n")

# Prediction
new_building = pd.DataFrame({'floors': [8], 'lateral_force_N': [214000]})
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raw_predictions = model.predict(new_building)[0]

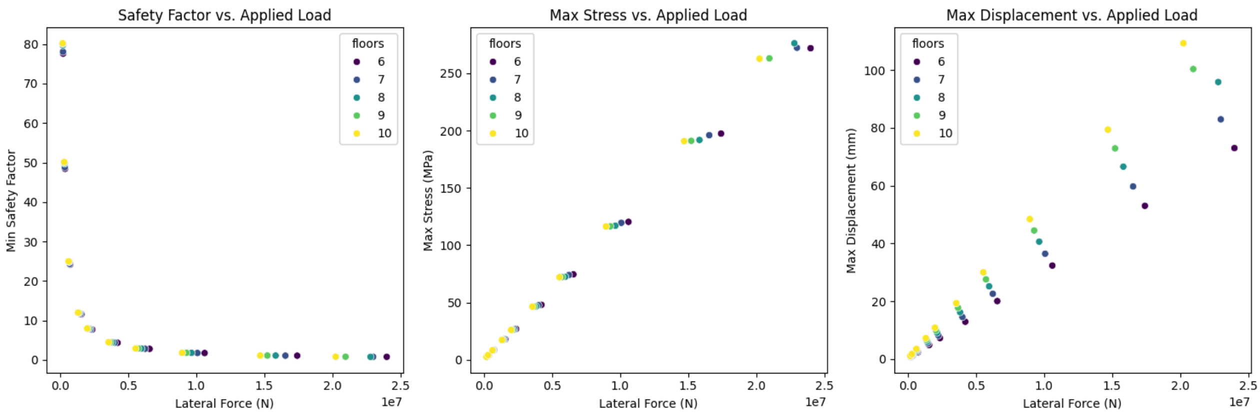
# Extract and invert the Safety Factor back to normal
pred_stress = raw_predictions[0]
pred_disp = raw_predictions[1]
pred_sf = 1 / raw_predictions[2]

print("--- Predictions for New Building (8 Floors, 214000 N) ---")
print(f"Predicted Max Stress:        {pred_stress:.2f} MPa")
print(f"Predicted Max Displacement:  {pred_disp:.2f} mm")
print(f"Predicted Safety Factor:     {pred_sf:.2f}\n")



# Safety Assessment
def assess_safety(safety_factor):
    """
    Standard engineering thresholds for building materials:
    < 1.0: Failing (Stress exceeds material yield strength)
    1.0 - 2.0: Marginal (Yielding may occur; little room for error)
    >= 2.0: Safe (Adequate buffer for unpredictable loads)
    """
    if safety_factor >= 2.0:
        return "Safe"
    elif 1.0 <= safety_factor < 2.0:
        return "Marginal"
    else:
        return "Unsafe"

# call the function
status = assess_safety(pred_sf)
print(f"Safety Assessment: {status}")
```




```
Model trained and saved as 'structural_ml_model_linear.pkl'

--- Model Evaluation (After Transformation Fix) ---
Max Stress:          R² = 0.9982, MAE = 2.8001 MPa
Max Displacement:    R² = 0.9676, MAE = 4.5546 mm
Min Safety Factor:   R² = 0.6925, MAE = 4.7515

--- Predictions for New Building (8 Floors, 214000 N) ---
Predicted Max Stress:        3.06 MPa
Predicted Max Displacement:  1.51 mm
Predicted Safety Factor:     67.77

Safety Assessment: Safe
```